

Uber: the dilemma of growing with disruptive innovation¹

Paulo Furquim de Azevedo
Leandro Simões Pongeluppe
Maria Clara de Azevedo Morgulis
Nobuiuki Costa Ito

It was Monday, August 3, 2015.² Daniel Mangabeira, director of government relations at Uber, a technology company that digitally connects passengers to drivers, had just hopped into an Uber car that he had requested four minutes earlier to take him to a meeting with government representatives at the company's offices in the center of São Paulo. A year after launching operations in the city, Uber had already grown significantly. Estimates pointed to a total of approximately 7,000 drivers and 700,000 users. This growth was exciting for the users and drivers joining the platform, but, on the other hand, was drawing the attention of

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² Fictitious example.

government authorities, who had yet to decide how to deal with this new business model and, most importantly, how to handle the issue raised by taxi drivers, who feared what they called "predatory competition." During the trip, Daniel Mangabeira took the opportunity to talk to the driver, Jorge Teixeira. As the conversation went on, it became more interesting. Jorge told Daniel that he had just recently become an Uber partner driver after leaving his job as a motorcycle courier and buying a vehicle that matched Uber's requirements, for the approximate amount of R\$70,000 supported by a loan. He said business was going well; on average, he was making R\$6,000.00 per month. However, despite the satisfactory income, Jorge seemed worried about the future of the business. After many various incidents between taxi drivers and Uber drivers, as well as the uncertainties regarding how the service would be regulated in São Paulo city, Jorge was afraid of not earning a return on his investment and being left to pay the loan on his car.

Uber's entry into the Brazilian market caused strong reactions from taxi drivers, who rallied support from local authorities in a bid to ban the app. Under the Uber business model, growth is imperative, because the more passengers and drivers that join the platform, the greater the value created for both. On the other hand, this growth incites reactions from those who may feel threatened by this new business model, a situation made worse by the lack of specific regulations for the activity. As such, one question kept troubling Mangabeira: how can you grow with a disruptive technology based on a network economy, but still subject to strong legal uncertainty?

The ride finally came to an end. Mangabeira got out of the car without even touching his wallet, since one of the benefits offered to Uber users is simple payment via their pre-registered credit card. However, as he was saying goodbye to Jorge, he thought about how the meeting at city hall would be important not only for Uber, but also for the lives of many drivers and users involved in the business.

Uber: birth and growth

Uber was born in the United States in San Francisco, California, in early 2009. Founded by Garret Camp and Travis Kalanick, the initial concept was to offer a luxury transportation service via a smartphone app. Operations in the testing phase were launched in 2010, but the commercial launch came only in 2011, initially limited to the city of San Francisco. Although a long time passed between the company's creation and launch, it expanded rapidly. Still in 2011, Uber already was operating in Paris and continued to expand in the United States.

Between 2012 and 2014, the company carried out an intense process of international expansion to several continents. In 2012, for instance, it already was present in Canada, United Kingdom and Australia. In 2013, Uber services became available in Asia and Africa, in countries

like Singapore and South Africa. The service became available in Latin America only in 2014, starting with Tijuana, Mexico. In October 2015, Uber was operating in more than 330 cities in 60 countries.

Geographic growth also was followed by expansion of the services offered by Uber to target various market segments. For the high-end segment, the company developed UberBLACK, with black sedan cars holding up to five passengers. UberX was created for the price-sensitive segment, using compact cars with significantly lower fares. UberSUV is also a luxury service, with large, black cars for users who require more space. In addition to these segments, the company adopted service portfolios and variations based on the demands of each city. In 2015, for example, there was UberEXEC and UberXL in London, and UberBERLINE in Paris. Service requirements also varied from place to place.

To use Uber's services, users install an app on their smartphones and register their name, mobile number, e-mail, language and credit card information. When they open the app, users see a map on which they can adjust their location. Through GPS, the Uber app will associate the soliciting user with the closest drivers and estimate a wait time. Users also can see the estimated fare of the trip before confirming their request. After confirming the user's exact location, the app contacts the other end of the platform: drivers. Once the driver accepts the trip, the passenger will receive information about the driver and car and can track them on the app's map (Appendix 1 – Image of the Uber application). The user is notified about the driver's arrival by text message.

Once in the car, passengers can see the service's quality standard. Upon arriving at their destination, passengers do not need to pay, since the fare is charged directly to their credit card registered in the app. The final fare also can be divided among more than one Uber user. At this stage, the driver and passenger rate their experience so that the app can collect information about users and drivers as a means to encourage and monitor the quality of the service provided. Partner drivers are free to exercise their activities as they wish. They are not subject to orders, are free to decide when and how long they will work and are not subject to exclusivity. Their interest in remaining on the platform and receiving more requests from passengers is what encourages them to seek good ratings, which in turn helps to ensure the service's quality standards. In Brazil, Uber retains 20% of the price of every trip by UberBLACK and 25% by uberX, with the remainder paid to the driver.

Around the world, Uber's entry into markets has faced strong resistance from government authorities, taxi drivers, taxi companies and trade unions. Many of the reactions can be explained by the novelty of the service and how rapidly it has entered markets. The lack of regulation and potential competition with a heavily regulated service (taxis) led to a situation in which Uber cars were operating more freely than taxi drivers. As such, Uber proved to be a

potential competitor for taxis without having to comply with the same regulations, which caused negative reactions.

The three main reactions against Uber were: (1) accusations of unfair competition, since Uber was not required to comply with the same regulations as taxi drivers; (2) concerns with safety, since the local government authorities did not inspect vehicles or the drivers connected to the platform; and (3) the work relationship between drivers and Uber, since Uber sees drivers as entrepreneurs who provide their service independently, while critics argue that an employment relationship exists that requires compliance with labor laws. Demonstrations and protests around the world led to physical confrontations, assaults and destruction of property. These conflicts led to the suspension of some services provided via the app by court decision in certain cities in Spain in December 2014³, and to two Uber executives being asked to provide clarifications in France in June 2015⁴, as well as to various other incidents and restrictions on the app's use around the world.

Disruptive innovation, network economy and status quo: seeking the ideal balance

Uber's business model is based on a tripod that involves major dilemmas. First of all, it represents a disruptive innovation, since it radically changed the relationship between people who would like to go somewhere and people who owned an idle vehicle and were willing to transport the former. When it was launched, Uber presented an economic opportunity by making the relationship between potential passengers and drivers easier, based on the exponential growth in the number of smartphones, which provided both sides with connectivity and georeferencing. Through Uber's digital platform, this transaction occurs directly, rapidly and safely, whether with regard to the fare and payment of the trip or to quality and reliability for passengers and drivers.

Uber mitigated the first risk by determining that payment would be made directly via credit card and that drivers would be paid via transfers to their bank account upon provision of the service. The second risk was mitigated by the driver validation method when they register in the system and based on user ratings.

The immediate result of this type of innovation was to reduce information asymmetry between the parties and consequently ensure a safer and more reliable transaction. This was the foundation for regulating taxi service, which managed only precariously to address this

³ Source: *BBC*. Available at: <<http://www.bbc.com/news/business-30395093>>. Accessed on: Oct. 13, 2015.

⁴ Source: *Time*. Available at: <<http://time.com/3940092/uber-executives-arrested/>>. Accessed on: Oct. 13, 2015.

informational problem. In the words of Vinícius Marques de Carvalho, president of Brazil's antitrust agency CADE, as expressed in an article published in the newspaper *Folha de S.Paulo* on July 12, 2015:

*Taxi service is characterized by information asymmetries. When you call a taxi, you don't know the conditions of the vehicle, the final price to be paid for the trip or the reliability of the driver.*⁵

Uber's business also is subject to network externalities, i.e. the higher the number of partner drivers connected to the platform, the higher the value created for potential passengers and vice-versa. This happens because as the number of partner drivers increases, the probability of having a driver available within the passenger's range is higher, reducing wait times. Similarly, the higher the number of users, the higher the benefit created for drivers, who will be able to do more trips and use their investment in the vehicle more efficiently. Therefore, the success of the business is due not only to the fact that it is a disruptive innovation, but also that it requires a rapid gain in critical mass (number of people connected to the platform) sufficient to make it viable.

To be profitable, Uber would have to grow until reaching the critical mass present in the early majority of users (Appendix 2) so as to make its business model feasible sustainably. The longer it takes the company to reach this critical mass, the higher the deficit of its operations, not to mention that such slow growth could scare off early adopters since they would see little benefit from joining a network that doesn't grow. As such, the feasibility of Uber's business depended on rapid development to reach critical mass and become sustainable.

However, by definition, every disruptive innovation affects existing social and economic structures. By promoting the rapid development of its business model in Brazilian cities, Uber started to change the status quo of the market for private urban transportation of individuals, affecting the public mode of this transport, which is represented by taxis licensed by local authorities. Virtually the only ones to provide this type of service until recently, taxi drivers are now facing competition from a new business model that operates under very different rules, including with regard to legal limits on new drivers. As expected, the reaction of taxi drivers to Uber was intense and should become even more intense the faster that this disruptive innovation gains market. On the other hand, for Uber, the feasibility of the business depends on rapid growth. What could be done?

Uber in Brazil: entry strategy and regulation

⁵ Source: *Folha de S.Paulo*. Available at: <<http://www1.folha.uol.com.br/mercado/2015/07/1654462-melhor-que-proibir-o-uber-e-fortalecer-os-taxistas-diz-chefe-do-cade.shtml>>. Accessed on: July 13 2015.

In Brazil, Uber launched its operations in Rio de Janeiro just before the FIFA World Cup, in May 2014. By June 2015, the company was already operating in four Brazilian cities: Rio de Janeiro (state of Rio de Janeiro), São Paulo (São Paulo), Belo Horizonte (Minas Gerais) and Brasília (Federal District), offering the services UberBLACK, UberX and UberBIKE. In general, Uber services in Brazil were being provided as described above, except for UberBIKE, which did not mean renting a bicycle, but the possibility of asking for a car that could carry a bike. This option allows users to take an Uber for part of a route that they will finish by bike at another time. The fares and conditions of the vehicles were the same as UberBLACK or uberX (Appendix 7 – Services available in each Brazilian city).

As expected for a business model based on network externalities, Uber's entry strategy in Brazil was to grow rapidly, concentrating its operations in regions with higher potential demand, followed by expansion into other regions of the same city; first Rio de Janeiro, then other capitals with high density and income. This strategy is justified by the nature of the business, which required rapidly building a mass of users (passengers) and service providers (drivers) to create network economies. So, in order to operate, the Uber platform must have available a large number of drivers that could immediately provide the trips requested. With that, users would feel encouraged to install and use the app. At the same time, if there were no users, drivers would be idle and wouldn't be encouraged to register. These mutually dependent factors justified the rapid entry strategy. However, with the same speed that Uber was expanding its operations in Brazilian cities, taxi drivers and local governments were reacting to its arrival.

The impact of Uber on taxi transportation

In the cities where Uber was operating, the immediate reaction came from taxi drivers, who were the exclusive providers of public transportation of individual passengers by car, establishing a type of market reserve. In fact, to operate a taxi, drivers required a permit from the city's authorities, which rarely handed out new permits. As an alternative, those interested in driving a taxi were able to buy a permit from working drivers, paying a higher entry cost.⁶ In exchange, drivers had the advantage of purchasing cars with exemption from taxes and of potentially earning more, since they were operating in a market where the number of service providers was limited by the government. Other drivers without the financial conditions to buy a permit on the parallel market opted to lease a permit and vehicle from a third party, paying a daily rate that includes the costs of the vehicle and the right to operate as a taxi driver.

⁶ This transaction was, however, illegal, since taxi permits are individual and non-transferable. Still, there is evidence of informal purchases and sales of permits.

When it established itself in the market, Uber broke this logic and not only served as an alternative means of transportation, but most importantly allowed new drivers to provide the service, including those already working in executive transport companies and those encouraged to give the job a try. Those most affected by this new business model were the people who held permits (whether they were drivers or not), whose value arose from the right to prohibit the entry of new drivers and therefore was assured supra-competitive earnings. For example, before Uber, the market had a pre-established number of taxi permits, which did not vary in accordance with fluctuations in supply and demand, but rather based on the local government's decision. The government, however, could subject its interests to those who already had permits and would not like to see competition grow. Also according to Vinícius Marques de Carvalho:

*Take the example of Brasília: it's been 30 years since a new taxi permit was issued. What was the population of Brasília then and what is it today? Do you think that demand hasn't grown?*⁷

In theory, these permits should not be traded in the market (although they often were (Appendix 3)), but they could be leased, through which the holder would receive an income as a result granting use rights. Appendix 4 provides an estimate of the net present value of a taxi permit based on a few economic parameters. Appendix 5 shows five years of cash flow for a taxi driver without a permit, excluding car costs and average daily fares in São Paulo.

For example, in Rio de Janeiro, the first Brazilian city where Uber began operating, a permit costs approximately R\$900,000, based on the average daily lease price for the permit and vehicle; on the average cost of a vehicle (R\$50,000.00) renewed every five years, deducting the tax exemptions for taxi drivers acquiring a vehicle; and on annual maintenance and insurance cost of approximately R\$4,000, resulting in a cash flow, from which is deducted the net present value based on the real interest rate of 4.22% p.a.⁸ (risk-free rate discounting inflation in the last 12 months).⁹ Meanwhile, in São Paulo, this amount, based on average daily lease price, a permit would cost approximately R\$680,000. This means that Uber's entry in the market should increase the supply of services to the point where there is no value in the right to limit third-party participation and potentially invalidating the value of the permits. By multiplying the net present value of each permit by the number of permits existing in each of these four cities, we

⁷ Ibid.

⁸ This amount is higher than the amount reported for permit purchase and sale transactions in Rio de Janeiro. It is, however, an informal transaction, which, as such, poses risks and therefore its negotiable value is lower than the income expected to be generated by the asset. Therefore, the calculation is based on the income generated from permit leasing.

⁹ Since the permit does not depreciate, the series could go to infinity. We opted for a more conservative cash flow of 30 years, with the purchase of a vehicle every five years.

arrive at a value of approximately R\$58 billion, arising from the power to prevent the entry of new competitors. This is the size of the resistance to Uber's expansion (Appendix 6).

This shows the complexity of the matter faced by Uber. If, on the one hand, it needed to expand rapidly to guarantee the benefits of network economies to its users, on the other hand, in rushing to build a critical mass of clients and suppliers it faced strong opposition from those who were benefitting from the previous status quo. The reactions to Uber's business model surpassed expectations. And their effects could go beyond mere street protests, since taxi drivers are known to hold a strong influence over local governments due to their long-term relationship.

On the other hand, users of the Uber system weren't sufficiently able to take collective action to defend the model. The individual gains of each user would be relatively small, limiting the ability and interest of coordination of so many people. Those who were adversely affected by the Uber model, particularly permit holders and taxi driver unions, had broad capacity to coordinate a collective action. As explained by professor João Manoel Pinho de Mello and Vinícius Carrasco in an article published in *Exame* magazine on August 10, 2015:

*Extraordinary income creates incentives for maintaining them at all costs. (...)Therefore, the interests of permit holders in maintaining the status quo surpasses the interest of consumers in changing it.*¹⁰

This capacity was notably exercised by enabling and maintaining the status quo and allowing taxi drivers to take various forms of actions, from protests and blocking streets (Appendix 8) to even exerting political pressure and, sometimes, physical coercion in an attempt to halt Uber's progress. Felipe Souza's article in the newspaper *Folha de S.Paulo* on June 19, 2015, gives an example of this situation in the statement by the president of the São Paulo Taxi Drivers and Tax Companies Employees Union):

*"We can't stop our members," "there will be deaths."*¹¹¹²

Antônio Raimundo Matias dos Santos, known as Ceará, continued:

*"I'd like to ask our noble congressmen to take the appropriate measures, otherwise we won't be able to contain our members."*¹³¹⁴

¹⁰ Source: *Exame*. Available at: <<http://exame.abril.com.br/rede-de-blogs/leis-da-oferta/2015/08/10/o-uber-e-a-natureza-da-regulamentacao/>>. Accessed on: Aug. 11, 2015.

¹¹ Source: *Folha de S.Paulo*. Available at: <<http://www1.folha.uol.com.br/cotidiano/2015/06/1644905-vai-ter-morte-diz-taxista-sobre-regulamentacao-do-aplicativo-uber.shtml>>. Accessed on: Jun. 20 2015.

¹² Since italics and indentation are enough to let readers know that they are reading a quote, we suggest removing the quotation marks: *We can't stop our members. There will be deaths.*

¹³ Ibid.

Combined with Uber's exponential growth and the threat posed by it, this reaction made everyone even more eager for the implementation of a regulatory model for the industry.

The Uber effect in market regulation

According to Frederico Meinberg Ceroy, president of the Brazilian Institute of Digital Law, Uber's entry strategy in Brazil had a price:

"Uber's sudden and anxious entry has a price. This price is the aversion of governments, who feel disrespected."

Uber's rapid entry and growth in cities had an adverse effect on regulation. First, Uber was able to serve anyone who had a smartphone, becoming a very similar replacement for taxis, at least for this user group. However, it wasn't exactly a taxi service, because drivers won't take passengers who hail them on the street, for instance. And registered drivers weren't entitled to the tax benefits provided to taxi drivers. In other words, the service provided by Uber could not be defined as public transportation of individuals, because it is not "open to the general public."

It was concluded that Uber operated in a segment of the legal market that didn't exist until then, but became economically viable due to the dissemination of smartphones with GPS technology and a technological platform that connected drivers and passengers. Under these terms, the company's entry into the market was transformational, and there were no regulations or prior experiences to guide government authorities. In addition to the undisputed lack of regulation, there were the many organized interests of those who exploited the taxi market, especially permit holders, and the dominant reaction was to vote for a municipal law prohibiting the app's use. For example, many Brazilian cities had draft bills to prohibit Uber even before the company started operating: Curitiba (state of Paraná), Vitória (Espírito Santo), Goiânia (Goiás), Joinville (Santa Catarina) and Manaus (Amazonas).

In the cities where Uber was operating, many different views and solutions were being discussed. In São Paulo, the mayor said he was in favor of regulating this new technology, going against the decision by the City Council, but the initial move was to restrict the app, with the application of fines and apprehension of vehicles. In Rio de Janeiro, the City Council already had passed a bill to ban Uber, which was sanctioned by the mayor, but the courts granted an injunction authorizing the company to operate. In Belo Horizonte, a bill sought to limit the app's

¹⁴ Since italics and indentation are enough to let readers know that they are reading a quote, we suggest removing the quotation marks: *I'd like to ask our noble congressmen to take the appropriate measures, otherwise we won't be able to contain our members.* [From now on, the suggestion to remove the quotation marks is implicit.]

activities to fulfilling demand not met by taxis. Rodrigo Rollemberg, the governor of the Federal District, vetoed the bill passed by the District Council of Brasília to prohibit the app, and submitted to the local legislative branch an innovative regulation to define locally the basis for rendering the service, but which is still pending approval by the District Council. As of the conclusion of this case study, the São Paulo city government was the only one to have presented, via a decree, a regulatory model for individual transportation contemplating services such as that provided by Uber. The scenario is marked by uncertainties, which directly affects the company's decisions, as well as the decisions of potential competitors.

Arriving at the office in São Paulo

The conversation with government authorities had been intense, and Daniel Mangabeira was worried when he walked into the meeting room at Uber's offices in São Paulo. The business model based on rapid entry into new markets was not consistent with awaiting the definition of specific regulations for the services or with entry limited to market niches that wouldn't disturb existing service providers. However, despite the benefits provided by a rapid gain in critical mass, this was the main source of adverse reactions faced by the company. Daniel Mangabeira was the main representative in contact with public authorities, playing a key role in Uber's future decisions: what growth strategies should Uber adopt in the coming years? Would it be possible to support its growth without eliciting strong opposition from taxi drivers?

Appendices

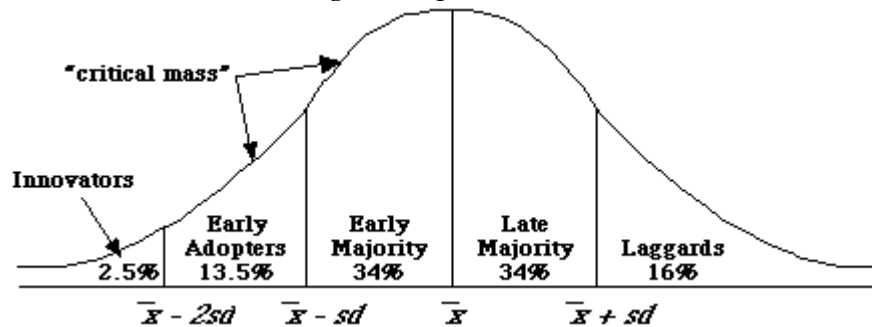
Appendix 1 - Uber's application and service



Note: On the right, a photo of the app when requesting a car. On the left, a photo of the uberX service.

Source: Uber.

Annex 2 – New technologies adoption curve



Source: E. Rogers (1995). *Diffusion of Innovations*. New York: Free Press.

Appendix 1 – Example of alternative form of selling taxi permits



Troco Apartamento Em Licença De Taxi **R\$ 200.000,00** Artigo usado Rio De Janeiro

Troco Apartamento Em Licença De Taxi

Troco este apartamento em um Táxi com autonomia. O valor médio do imóvel gira em torno de R\$ 200.000,00. O aluguel deste apartamento está em torno de R\$ 1.200,00. O prédio fica num lugar muito bonito e muito seguro, no condomínio Jardim Clarice no Anil, em Jacarepaguá.

Source: Mercado Livre. Available at: <<http://apartamento.mercadolivre.com.br/MLB-705471096-troco-apartamento-em-licenca-de-taxi-JM>>. Accessed on: Oct. 6, 2015.

Appendix 2 – Estimated NPV of a taxi permit in the cities where Uber operates

City	Average daily rate 8h (R\$/day)*	NPV of taxi permit	Average cost of vehicle (R\$)	Number of days in year (days)	Average discount on IPI + ICMS taxes for taxi drivers (%) **	Real interest rate (% p.a.) ***
Belo Horizonte	R\$140.00	R\$505,071.50	R\$50,000.00	365	30.00%	4.22%
Brasília	R\$200.00	R\$811,363.57				
Rio de Janeiro	R\$220.00	R\$913,460.93				
São Paulo	R\$175.00	R\$683,741.87				

* Sources: *Folha*; Acat.taxi.

** Discount may vary from state to state.

*** Source: Central Bank of Brazil. Available at: <<http://www.bcb.gov.br/htms/selic/selicdia.asp>>. Accessed on: Sept. 23, 2015. Excluding inflation of 9.53% in 12 months.

**** Source: Brazil's Federal Revenue Service.

Appendix 3 – Cash flow of a taxi driver in São Paulo

Annual depreciation legal rate (%)*	Average number of business days per year (days)**	Average daily rate 8h in São Paulo (R\$/day)***	Average daily fuel cost 8h (R\$/day)	Interest rate**** (% p.a.)	Interest rate (daily factor)****	Net taxi revenue per hour (R\$/h)	
20%	253	R\$175.00	R\$60.00	14,15%	0.0100052531%	R\$31.25	
Average cost of vehicle (R\$)	Average discount on IPI + ICMS taxes for taxi drivers (%) ****	Average cost of vehicle for taxi drivers (R\$)	Approximate annual expense (average daily rate and fuel) (R\$)	Average hours worked per day (h)	Approximate annual revenue	Annual amortization (R\$)	Approximate annual NPV (5 years)
R\$50,000.00	30%	R\$35,000.00	R\$59,455.00	8	R\$63,250.00	R\$7,000.00	R\$6,265.48
			R\$118,910.00	16	R\$126,500.00	R\$7,000.00	R\$19,247.19
			R\$178,365.00	24	R\$189,750.00	R\$7,000.00	R\$32,228.90

* Source: Brazil's Federal Revenue Service.

** Excluding weekends and holidays.

*** Sources: Sebrae; G1; Acat.taxi.

**** Source: Central Bank of Brazil. Available at: <<http://www.bcb.gov.br/htms/selic/selicdia.asp>>. Accessed on: Sept. 23, 2015.

**** Discount may vary from state to state.

Appendix 4 – total NPV of the permits

City	Number of permits	NPV of permit (R\$)	NPV in permit/city (R\$ '000)
Belo Horizonte	6,992	R\$ 505,071.50	R\$ 3,531,459.92
Brasília	3,400	R\$ 811,363.57	R\$ 2,758,636.14
Rio de Janeiro	32,000	R\$ 913,460.93	R\$ 29,230,749.70
São Paulo	34,000	R\$ 683,741.87	R\$ 23,247,223.72
Total			R\$ 58,768,069.48

Sources:

São Paulo city government. Available at: <www.prefeitura.sp.gov.br/cidade/secretarias/transportes>. Accessed on: Oct. 12, 2015.

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G1. Available at: <g1.globo.com/distrito-federal/noticia/2015/07/apos-acordo-gdf-anuncia-700-novas-autorizacoes-imediatas-para-taxistas.html>. Accessed on: Oct. 12, 2015.

Belo Horizonte City Council. Available at: <www.cmbh.mg.gov.br/chapeu/servico-de-taxi>. Accessed on: Oct. 12, 2015.

Appendix 5 – Uber services available in Brazilian cities

City	uberX	UberBLACK	UberBIKE
Rio de Janeiro (Rio de Janeiro)	Yes	Yes	No
São Paulo (São Paulo)	Yes	Yes	Yes
Belo Horizonte (Minas Gerais)	Yes	Yes	No
Brasília (Federal District)	Yes	Yes	No

Source: Uber's website. Accessed on: Oct. 15, 2015.

Appendix 8 – Taxi drivers protest against Uber



Source: G1 Portal. Available at: <<http://g1.globo.com/sao-paulo/noticia/2015/09/manifestacao-de-taxistas-contr-uber-trava-ruas-importantes-do-centro-de-sp.html>>. Accessed on: Oct. 6, 2015. Photos: Carolina Dantas/G1.